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EDUCATION

NYU Tandon School of Engineering **Expected May 2027**
Master's in Computer Science *New York, United States*
Coursework: Analysis of Algorithms, Distributed Systems, Principles of Database Systems, Machine Learning

WORK EXPERIENCE

Software Developer Intern **Sept 2023 - Dec 2023**
Enabled Analytics *On-Site*

- Developed a custom Salesforce-based order management system integrated with WhatsApp, allowing supermarket customers to browse items, place orders, and receive automated responses through a chat-based workflow.
- Built Apex, SOQL/SOSL, and Salesforce Flow automation to process WhatsApp orders in real time, validate item availability, update inventory, prevent duplicate orders, and sync order data across Salesforce records.
- Created Salesforce Lightning interfaces for supermarket staff to monitor WhatsApp orders, update fulfillment status, manage inventory, and access customer/order history from a centralized dashboard, owning the feature end-to-end from design through testing and production deployment while debugging workflow-execution issues across order placement and inventory sync.

PROJECTS

Frontier - Real-Time Research Intelligence & Scholarly Knowledge Graph Platform | *Python, FastAPI, Kafka, PostgreSQL, Redis, OpenSearch, OpenTelemetry, Docker, Kubernetes*

- Built a distributed research-intelligence platform ingesting OpenAlex and Crossref metadata into a temporal, schema-flexible knowledge graph of papers, authors, institutions, topics, and citations; engineered Kafka-based entity-resolution, deduplication, incremental-sync, and historical-backfill pipelines with priority scheduling, backpressure, and transactional event propagation.
- Engineered a high-performance PostgreSQL graph/query layer using JSONB, GIN/composite indexes, recursive CTEs, materialized relationships, and RLS-based inherited workspace permissions; implemented hybrid graph/semantic retrieval for cited LLM research synthesis and trend discovery, exposed through FastAPI with Redis caching, OpenSearch, and OpenTelemetry observability.

P2P Stock Trading Engine | *Python, WebSockets, Kafka, Redis, PostgreSQL*

- Built an event-driven, real-time trading engine with market-data ingestion, WebSocket fan-out, a price-time-priority limit order book, and deterministic settlement, backed by per-contract in-memory matching services and an append-only PostgreSQL ledger kept consistent through idempotency keys and unique constraints.
- Profiled the critical matching path under a simulated burst-traffic load-testing harness, diagnosed in-memory versus durable state desync failure modes, and reduced p99 order-processing latency by 10x while preserving deterministic execution under high-throughput concurrency.

Recall - Multimodal Personal Search & Memory Platform | *Python, FastAPI, PostgreSQL, OpenSearch, Amazon SQS, Redis, Docker*

- Built a cloud-native multimodal search platform with an event-driven ingestion pipeline over Amazon SQS, where independently scaling workers process files by queue depth using idempotent operations, retry policies, and dead-letter queues to stay consistent under duplicate delivery and worker failures.
- Engineered hybrid retrieval over PostgreSQL and OpenSearch that fuses full-text, vector, and metadata filtering with reranking, enforced strict per-user tenant isolation across the relational and search layers, and lazily promoted files to deeper indexing to balance real-time queries against expensive batch processing.

SecurePR - Secure PR Execution and Verification Platform | *Python, FastAPI, Temporal, Kubernetes (GKE), gVisor, seccomp, Cilium*

- Architected a zero-trust platform that sandboxes and executes untrusted GitHub pull-request code end to end on Linux isolation primitives - unshare()-based PID, mount, UTS, and IPC namespaces, dropped capabilities, and empirically derived seccomp allowlists - then reports verdicts through the GitHub Checks API.
- Orchestrated per-PR execution as durable Temporal workflows scheduling Kubernetes Jobs across kind and GKE clusters, root-causing production failures spanning cgroup versions, queue passthrough, and missing syscalls by instrumenting real workload traces instead of guessing at configuration.

RESEARCH AND PUBLICATIONS

Effective Analysis of Machine and Deep Learning Methods for Diagnosing Mental Health Using Social Media Conversations
IEEE Transactions on Computational Social Systems

- Analyzed 100K+ tweets and developed NLP pipelines using BERT, RoBERTa, DistilBERT, XLNet, and classical/deep-learning models for behavioral risk-pattern detection.
- Evaluated model accuracy and sentiment features to build predictive systems that identified psychological-condition signals with 97% accuracy.

SKILLS

Languages and Frameworks: Python, TypeScript/JavaScript, SQL, FastAPI, Node.js, React, Next.js, data structures and algorithms
Databases and Data Modeling: PostgreSQL (JSONB, indexing, recursive CTEs, row-level security), ClickHouse, Redis, OpenSearch, query optimization
Distributed Systems and Infrastructure: Kafka, Amazon SQS, event-driven pipelines, idempotency, dead-letter queues, backpressure, Temporal, Airflow, Spark Structured Streaming, Docker, Kubernetes
Systems, Security and Observability: Linux isolation (namespaces, cgroups, seccomp), gVisor, Cilium, load testing, latency profiling, OpenTelemetry, Datadog, CI/CD, unit/integration testing
Certifications: Machine Learning Specialization by Andrew Ng, Google Cloud Digital Leader